CS2103 / Lab – Assignment 05

Topic: Logistic Regression and K-Nearest Neighbors (KNN)

Instructions: Complete all five questions. Implement **Logistic Regression and KNN from scratch using NumPy**. Built-in classifiers and optimizers are not allowed.

Datasets: UCI Banknote Authentication UCI Heart Disease

Q1. Logistic Regression from Scratch

Using the **Banknote Authentication Dataset**, implement Logistic Regression with

$$z = \mathbf{w}^T\mathbf{x} + b, \qquad \sigma(z) = \frac{1}{1+e^{-z}}.$$

Implement BCE loss, probability prediction, and classification using $\tau = 0.5$. Report the final training/test loss and percentage of correctly classified test samples.

Q2. Gradient Descent

For

$$J = -\frac{1}{m}\sum_{i=1}^{m}\left[y_i \log(p_i) + (1-y_i)\log(1-p_i)\right],$$

derive $\partial J/\partial \mathbf{w}$ and $\partial J/\partial b$. Implement Batch Gradient Descent and plot **BCE Loss vs. Training Iterations** to demonstrate convergence.

Q3. Learning-Rate Analysis

Train the model using

$$\alpha \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}.$$

Compare the loss curves and identify cases of **slow convergence, stable convergence, oscillation, and divergence**. Briefly explain the observed behavior.

Q4. KNN from Scratch

Using the **Heart Disease Dataset**, implement KNN using Euclidean distance

$$d(\mathbf{x}, \mathbf{x}_i) = \sqrt{\sum_j (x_j - x_{ij})^2}.$$

Test $k \in \{1, 3, 5, 7, 9, 15\}$. Report the percentage of correctly classified samples for each k, identify the best k, and explain the effect of very small and very large values of k.

Q5. Nonlinear Decision-Boundary Challenge

Generate 2000 samples with $x_1, x_2 \sim U(-2, 2)$ and

$$y = \begin{cases} 1, & x_1^2 + x_2^2 > 2, \\ 0, & \text{otherwise.} \end{cases}$$

Compare the decision boundaries obtained using:

(i) Logistic Regression with $[x_1, x_2]$,
(ii) Logistic Regression with $[x_1, x_2, x_1^2, x_2^2, x_1x_2]$,
(iii) KNN with $[x_1, x_2]$.

Briefly explain why standard Logistic Regression struggles with the nonlinear boundary and why KNN can model it without explicit feature transformation.

Note: Students must be able to explain the mathematical formulation, implementation, convergence behavior, effect of k, and decision boundaries during the evaluation/viva.

Bonus: Bonus marks will be awarded for the effective use of **Claude Code or VS Code prompts**.